

THEME ONE : SYSTEMS

UNIT

1

Interactions of Organisms



Get Started

What I Already Know



- Plants are found everywhere around us.
- There are some basic needs that plants depend on to grow up and survive such as :

- Air. - Water. - Sunlight.

- The opposite pictures show two potted plants :

Plant (A) has green leaves and grows well, while plant (B) is wilted and has yellow leaves.



Plant (A)



Plant (B)

- Plant (B) cannot grow well and die due to one or all of the following reasons :

- It may be placed in a dark place, so it doesn't get sunlight.
- It may not be watered regularly.
- It may be placed in a bad aerated area, so it doesn't get enough fresh and clean air.

- In this unit, you are going to study :

- How plants use sunlight, air and water to make their own food.
- Types of living organisms : producers, consumers and decomposers.
- The interaction between living organisms to get their needed energy through what is called "Food Chains" and "Food Web".

Example : The **squirrel** eats leaves, fruits, insects and lizards to get the energy it needs.

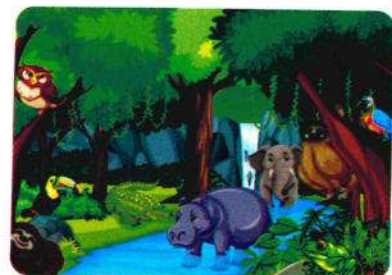
- What happens to an ecosystem, if a food chain in this ecosystem is interrupted.



Squirrel

- Unit Project : "Build a Miniature Ecosystem"

At the end of this unit, you are going to build a miniature ecosystem (small ecosystem) to show how living organisms depend on other living organisms to get their food. Also, the importance of some non-living things such as air, water, soil ... etc. for the survival in an ecosystem.



Ecosystem

CONCEPT

1.1

Plant Needs





Learning outcomes

By the end of this concept, your child will be able to :

- Use evidence to argue that plants use specialized structures to obtain the materials that they need to grow from Sun, air and water.
- Develop a model of how energy moves through plants.
- Develop a model of plant processes that use natural resources to complete life processes.
- Compare the structure and function of the vascular system in plants with the circulatory system in humans.

Key vocabulary

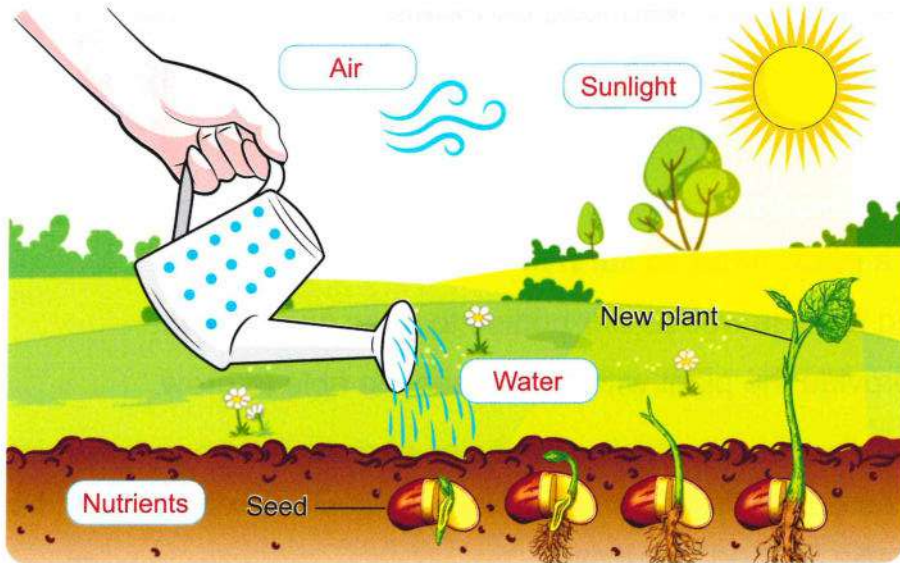
- Arteries
- Photosynthesis
- Vessels
- Circulatory system
- Plant
- Xylem
- Digestive system
- Stem
- Seed dispersal
- Stomata
- Germinate
- Survive
- Glucose
- Vascular system
- Nutrients
- Phloem
- Veins

Notes For Parents On Concept [1.1]

Lessons	Activities	What you should do with your child
1	Activity 1	Explain to your child how the structures of a plant use water, air and light to perform life processes.
	Activity 2	Discuss with your child what the plant needs to grow and survive.
	Activity 3	Discuss with your child basic and not basic plant needs for photosynthesis process.
2	Activity 4	Help your child germinate some seeds in a wet paper towel, then compare their growth to the growth of the other seeds which are placed in soil.
	Activity 5	Help your child do an experiment to show the effect of sunlight on plant growth.
3	Activity 6	Discuss with your child parts of a plant.
	Activity 7	Help your child do an experiment to observe how water and nutrients move from the roots to the leaves of a plant.
4	Activity 8	Let your child compare between the plant transport system and the human circulatory system.
	Activity 9	Discuss with your child how plants make their own food.
	Activity 10	Discuss with your child the function of flowers of plants.
5	Activity 11	Help your child to think about ways of seed dispersal in nature.
	Activity 12	Help your child to think like a scientist by answering a question about one of the main points of this concept, then write his/her claim, evidence and the scientific explanation.

LESSON ONE

Activity 1 Can You Explain ?



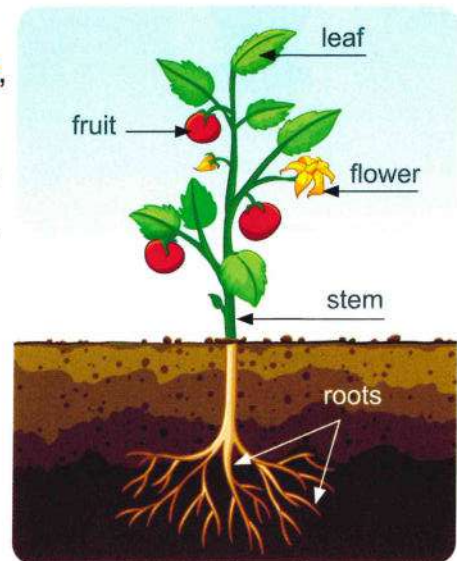
- When you observe the figure above that shows the steps of growing up a bean seed to form a new plant, you can find out what the plant needs to grow.
- Plants need **water**, **air**, **sunlight**, **nutrients** and **space** to grow.

► How do the structures of a plant use water, air and light to perform life processes ?

- Plants consist of different parts which are **roots**, **stems**, **leaves** and sometimes **flowers** or **fruits**, where each part of a plant has its own function.
- Plants use these different parts to obtain their **basic needs** of water, air and light to make their own food.

► In this concept, we will study :

- Plant basic needs.
- Parts of a plant.
- Comparing plant and human systems.
- Human circulatory system.
- Plant transport system.
- Plant food.
- Flowers and seeds.
- Seed dispersal.



Plant parts

bean seed
nutrients
perform

بذرة فول
العناصر الغذائية
يقوم

space
obtain
basic needs

مساحة
يحصل على
احتياجات أساسية

circulatory system
transport system
dispersal

الجهاز الدوري
نظام النقل
نثر / انتشار

Activity 2 Tree Needs

► Look at the opposite picture, then put (✓) or (x) :

- Both human and plant need food and water everyday to survive. ()
- Both human and plant need carbon dioxide gas to breathe. ()



► What does a plant need to survive ?

- Plants need food as well as our bodies to grow and survive.
- We must provide the plant with all its needs to able to grow.

Example : When a tree is planted, it begins to grow from a seedling into a mature tree depending on some resources such as **water**, **air** and **sunlight** to make its food to survive.

 Check your understanding

► Circle the items that the plant needs to grow and survive :



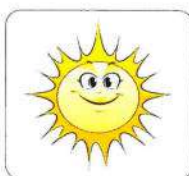
Space



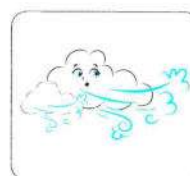
Water



Meat



Sunlight



Air



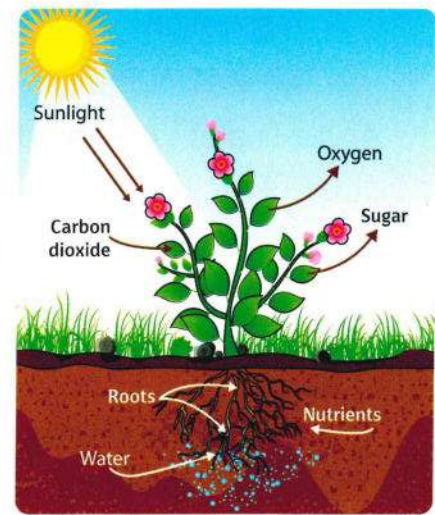
Perfume

Activity 3 What Do You Already Know About Plant Needs ?

- Water and air are basic needs of plants, animals and humans.
- Humans and animals need to eat food to gain energy and nutrients to live and grow.
- Most plants get nutrients and water from the soil and make their own food through a process known as "**photosynthesis process**" that takes place in the plant leaves.
- **Plants need some resources to live and grow such as :**
 - Sunlight. - Water. - Nutrients from the soil.
 - Carbon dioxide gas (needed for photosynthesis in plants).
 - Oxygen gas (needed for respiration of plants).

Plants and food

- Plants make their own food which is a type of **sugar** that provides the plant with energy to grow.
- Plants make their food (sugar) in their **leaves** by means of **photosynthesis process**, where :
 - The roots of a plant absorb water and nutrients from the soil.
 - Water and nutrients are carried from the roots to the leaves through the stem.
- **From the previous explanation, we can conclude that the plant's basic needs that enable it to make its food are :**
 - Sunlight. - Water. - Air (carbon dioxide).



Photosynthesis process



Check your understanding

- Classify the following items into "**Basic plant need for photosynthesis**" or "**Not basic plant need for photosynthesis**" :

(Water – Sunlight – Oxygen – Sugar – A forest – Carbon dioxide)

Basic plant need for photosynthesis	Not basic plant need for photosynthesis
.....	
.....	
.....	

In the Assessment Book :

Try to answer :

Self-Assessment ①

basic
gain
energy

أساسي
يكتسب
الطاقة

by means of
photosynthesis process
carbon dioxide gas

عن طريق
عملية البناء الضوئي
غاز ثاني أكسيد الكربون

absorb
forest
provide

يمتص
غابة
يمد

Exercises on Lesson 1

● Understand

● Apply

● Higher Thinking Skills

1 Choose the correct answer :

- 1. The of plant absorb water and nutrients from the soil. (Giza 2025)
a. roots b. stems c. leaves d. flowers
- 2. Humans and animals need to eat to get
a. oxygen gas. b. energy.
c. carbon dioxide gas. d. soil.
- 3. Plants make their food by a process known as (Alex. 2023)
a. respiration. b. absorption.
c. photosynthesis. d. digestion.
- 4. and are from the plant needs that help it make photosynthesis process. (Cairo 2023)
a. Oxygen – water b. Sunlight – carbon dioxide
c. Water – earthworms d. Nutrients – oxygen
- 5. Plants and humans are similar in some of their basic needs to survive such as
a. sunlight and rocks. b. water and air.
c. carbon dioxide and soil. d. soil and water.
- 6. Plants take from the air to make its food. (Alex. 2024)
a. water b. oxygen gas
c. carbon dioxide gas d. sugar
- 7. All the following are plant basic needs to make its own food, except
a. water. b. air. c. sunlight. d. rocks.
- 8. Which of the following sentences is wrong ?
a. Plants need sunlight to grow.
b. Plants roots absorb water from the soil.
c. Plants make their own food by respiration process.
d. Plants make their own food in their leaves.
- 9. Water and nutrients are carried from the roots to the leaves through the (Giza 2025)
a. stem. b. soil. c. fruits. d. flowers.
- 10. In photosynthesis process, plant produces to get energy.
a. oxygen gas b. sugar
c. carbon dioxide d. water

2 Choose from column (B) what suits it in column (A) :

(A)	(B)
1. Sunlight	a. is absorbed by the roots of the plant.
2. Carbon dioxide	b. is necessary for plant's growth.
3. Water	c. is not a basic need for plant growth.
4. Oxygen	d. is a gas which is produced during photosynthesis process.
	e. is a gas which is used by the plant during photosynthesis process.

1.

2.

3.

4.

3 Put (✓) or (X) :

- 1. Plants need water and air only to grow. (Menoufia 2023) ()
- 2. Stem of the plant absorbs water from the soil. (Cairo 2025) ()
- 3. Human, animals and plants need food and water to survive. ()
- 4. Plants use the sunlight to make their own food. ()
- 5. Carbon dioxide gas is one of the plant needs that helps it to grow and survive. ()
- 6. Photosynthesis process takes place in the plant roots. (Minia 2025) ()
- 7. The plant can make its own food in the absence of water. ()

4 Complete the following sentences :

- 1. Plants consist of stem, and
- 2. Plants absorb and from the soil through their
- 3. Plants make their own food through process that takes place in their
- 4. The stem carries water and nutrients from the to the of the plant. (Luxor 2023)
- 5. The plants use the light of to make their own food.
- 6. The food of plants is a type of which is made in their by photosynthesis process.
- 7. Soil is the source of and nutrients which the plant needs to make its own food.

5 Write the scientific term of each of the following :

- 1. A gas taken from the air by leaves to help the plant to make its own food. (Dakahlia 2025) (.....)
- 2. A liquid substance that plants, animals and human need to survive. (.....)

- 3. A part of the plant that carries water and nutrients from the roots to the leaves. (Damietta 2025) (.....)
- 4. The process by which the plant can make its own food. (.....)
- 5. The gas which is released from plants during photosynthesis process. (Ismailia 2025) (.....)

6 Cross out the odd word :

- 1. Carbon dioxide gas – Water – Oxygen gas – Nutrients. (Giza 2023) (.....)
- 2. Roots – Stems – Leaves – Sunlight. (.....)

7 Give reasons for :

- 1. Roots have important role in photosynthesis process of plants. (Sohag 2024)
.....
.....
- 2. Photosynthesis process is important for plants to survive. (Cairo 2023)
.....
.....

8 What happens if ...?

- 1. Plants have no stems.
.....
- 2. Plants can't get carbon dioxide gas from air. (Alex. 2025)
.....
- 3. We put a green plant in a dark room for many days. (Cairo 2024)
.....

9 Adam planted a flowering plant in a pot, He put this plant in a soil rich in nutrients and watered it everyday, he used to cover this pot everyday with a carton box to hide it from his brother, after many days, do you think that this plant will survive ? And why ?

- a. Yes, because it has nutrients and water.
- b. No, because it needs air and light.
- c. No, because plant doesn't need water and soil.
- d. Yes, because it can survive without sunlight.



LESSON TWO

Activity 4 Do Plants Need Soil ?

► Look at the opposite picture, then put (✓) or (x) :

1. Plants need air and sunlight only to grow. ()
2. If the plant is not watered for a long time, it will die. ()



Note

Germination means that the plant sprouts and begins to grow from a seed.

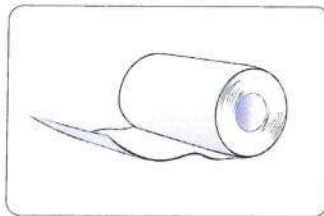
Do plants need soil to grow ?

To know whether plants need soil as a basic need for growth or not, we will germinate some seeds in a wet paper towel and measure their growth, and then compare their growth to the growth of the other seeds which are placed in the soil.

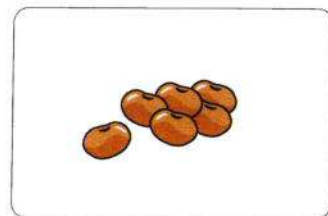
Tools



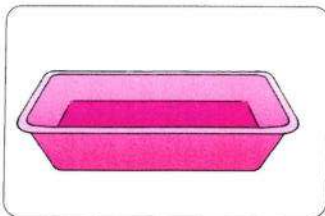
Plastic cup contains soil



Paper towels



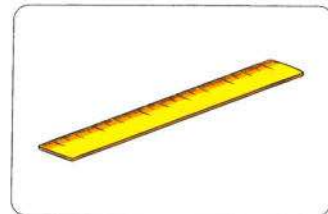
Six bean seeds
(Fava beans)



Plastic plate



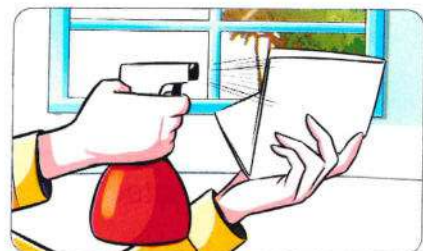
Water



Metric ruler

Steps

1. Use the water to wet the paper towel.



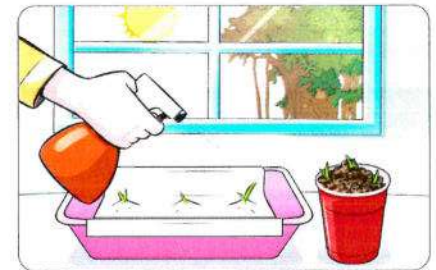
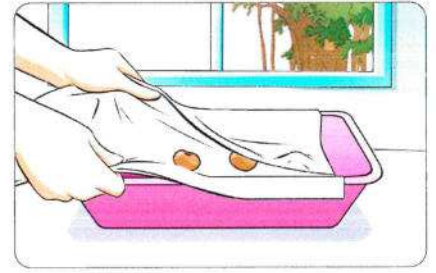
germination
wet
paper towel

إنبات
مبلل
منشفة ورقية
measure
growth
sprout

يقيس
نمو
ينبت
fava beans
metric ruler

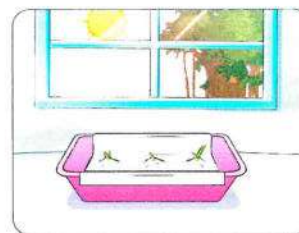
القول
مسطرة مترية

2. Place three seeds in the top half of the paper towel and fold the bottom half of the towel up so that it covers the seeds, then place the paper towel inside the plastic plate.
3. Plant the other three seeds in the cup that contains soil, then water the seeds.
4. Place the plate and the cup in a place where they can get sunlight.
5. Check the growth of seeds over the next several days. Wet the paper towel and water the soil as needed.
6. Measure the growth of each seed using the metric ruler.



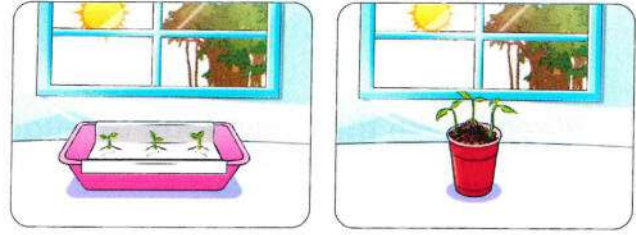
► Observations

- The initial growth of the seeds placed in the paper towel is similar to that of the seeds planted in the soil.



After 7 days

- The seeds grown without soil would not grow as quickly as the seeds in the soil.



After 14 days

► Conclusions

- In the presence of water, seeds can grow (germinate) without soil.
- In the presence of water and sunlight, plants can grow without soil for a while, but finally they need soil that provides plants with nutrients that allow plants to grow well.

? Give a reason for :

Plants can grow without soil for a while, but finally they need soil.

Because the soil provides plants with nutrients that allow plants to grow well.

Check your understanding

► Put (✓) or (x) :

1. The presence of soil is necessary for seeds in their initial growth. ()
2. When bean seeds grow in a wet paper towel, they need soil after a while. ()

Activity 5 Sunlight : A Basic Need

- Plants make their own food through **photosynthesis process**.

Photosynthesis process :

It is the process through which the green parts of plants (leaves) absorb sunlight to make their own food.

How can plants make their own food through photosynthesis process ?

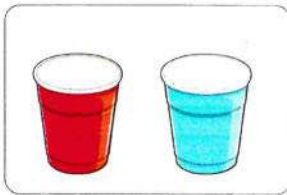
- Green plants use their leaves to collect sunlight and carbon dioxide from the air.
- Plant roots absorb water from the soil.
- Inside the green plants, sunlight allows carbon dioxide to combine with water to produce :
 - Oxygen** which is released in the air to help living organisms breathe.
 - Sugar** (the food of plant) which gives the plant the energy it needs to grow.

So, photosynthesis process can be represented as follows :

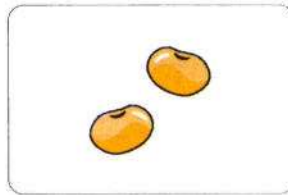
Sunlight + Carbon dioxide + Water $\longrightarrow$ Oxygen + Sugar

Now, we will do an experiment to show the effect of sunlight on plant growth :

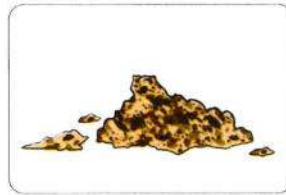
Tools



Red and blue plastic cups



Two bean seeds



Soil



Water

Steps

- Add the soil to the two cups, then put the bean seeds on the soil, where each cup contains one seed and cover the seeds with about 2 centimeters of soil.



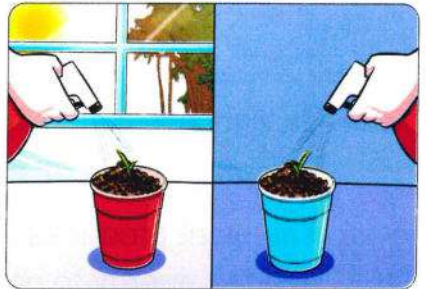
2. Add the same amount of water to each cup to moisten the soil.



3. Put the red cup facing the sunlight and the blue cup in a dark place.



4. Water both plants regularly and observe them along two weeks.



► Observations

After two weeks, we observe that :

- The plant in the red cup grew taller than the plant in the blue cup.
- The plant in the red cup has four leaves with dark green color, while the plant in the blue cup has two small leaves with pale green (yellow) color.

► Conclusions

- **Sunlight** is a basic need for the plants like water and air.
- **Sunlight** is important to plant growth, because plants use sunlight to make their own food, so the plant without sunlight does not grow well because it had less food.

In the Assessment Book :

Try to answer :

Self-Assessment (2)

Check your understanding

► Put (✓) or (x) :

1. In the presence of sunlight, plants can make their own food. ()
2. When a plant grew in sunlight, its leaves become pale green. ()

Exercises on Lesson 2

- Understand

● **Apply**

- Higher Thinking Skills

1 Choose the correct answer :

1. When the plant seed begins to grow and makes sprouts, this process is called
a. respiration. b. germination. c. absorption. d. reproduction.
2. If we put a bean seed in a, it may germinate.
a. dry paper towel b. wet paper towel
c. plastic plate d. dry soil
3. In the presence of water, seeds can germinate at the beginning of growth without the need of
a. soil. b. rocks. c. insects. d. dry paper towel.
4. Sunlight and carbon dioxide gas are collected by plant's to make its own food.
a. roots b. stems c. leaves d. flowers (Luxor 2025)
5. The plant produces through photosynthesis process that gives it the needed energy to grow. (Cairo 2023)
a. oxygen gas b. water
c. carbon dioxide gas d. sugar
6. Without, the plants can't grow well. (Suez 2023)
a. insects b. rocks c. sunlight d. moonlight
7. The roots of a plant absorb from the soil to help it grow.
a. oxygen gas b. carbon dioxide gas
c. sugar d. water

2 Put (✓) or (X) :

1. At the beginning of germinating some bean seeds, they can grow without soil and water. ()
2. All seeds need soil in its initial growth. ()
3. After many days, the growth of plant's seeds in a pot containing soil is similar to the growth of plant's seeds in a wet paper towel. ()
4. A green plant can grow well if it is placed in a dark room for many days. (Giza 2023) ()
5. Leaves of plants collect sunlight and carbon dioxide gas from air. ()
6. When the plant makes photosynthesis process, its leaves become weak and yellow. ()

- 7. Water and carbon dioxide are absorbed by plant's roots to help the plant to grow. ()

3 Correct the underlined words :

1. Respiration process helps the plant to make its own food. (.....)
2. Oxygen gas is absorbed by plant's leaves to make photosynthesis process. (Beni Suef 2023) (.....)
3. When a plant is placed in sunlight, its leaves become pale green. (.....)
4. Plant's leaves absorb water and nutrients from the soil. (.....)

4 Write the scientific term of each of the following :

1. The process by which plants make their own food using the energy of sunlight. (Aswan 2023) (.....)
2. Parts of the plant where sunlight allows carbon dioxide to combine with water during photosynthesis process. (.....)
3. A gas produced during photosynthesis process and it is needed for respiration of living organisms. (Kafir El-Sheikh 2025) (.....)
4. A substance that is produced from the plant during photosynthesis process and provides it with its needed energy. (.....)

5 Complete the following sentences :

1. In photosynthesis process, green plant gets from air to make its own food and produces that helps us to breathe.
2. Inside the green plant, sunlight allows carbon dioxide to combine with that is absorbed from the soil by plant's (Behira 2023)
3. The sugar that is produced from photosynthesis process provides the plant with it needs to grow.
4. The presence of , and air is very important for plants to grow.

6 Give a reason for :

- Green plants can make their own food.

.....

.....

7 What happens if ...?

1. We put a seed of bean in wet soil for many days.

(Giza 2023)

2. We put a bean seed in a wet paper towel for more than two months.

3. A plant is placed in a dark place for many days.

(Alex. 2025)

8 What is meant by (Define) :

Photosynthesis process :

9 Look at the following figures, then complete the following sentences using the words below :

(soil – figure (A) – figure (B))



Figure (A)



Figure (B)

- The seeds in grow faster than those in
- Seeds in figure (B) should be transferred into to complete its growth.

LESSON THREE

Activity 6 Parts of A plant

► Put (✓) or (x) :

1. The main parts of the plants are roots, stems, leaves and soil. ()
2. Each part of the plant has its own function. ()

- In this activity, we will study different plant parts in details.

Roots

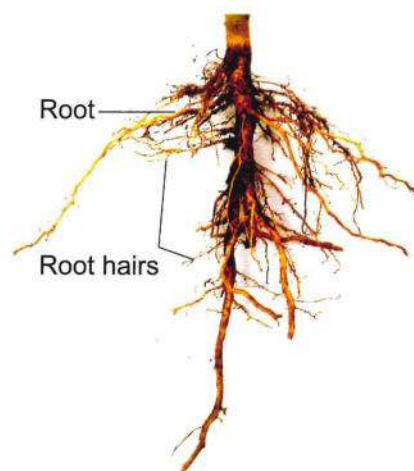
• Functions of the plant roots :

- Roots fix (anchor) the plant in the soil.
- Roots absorb (draw) water and nutrients from the soil to the plant.

- Plant roots have hairlike features (structures) called **root hairs**.

• Function of the root hairs :

- Root hairs absorb more amount of water and nutrients that the plant needs from the soil.



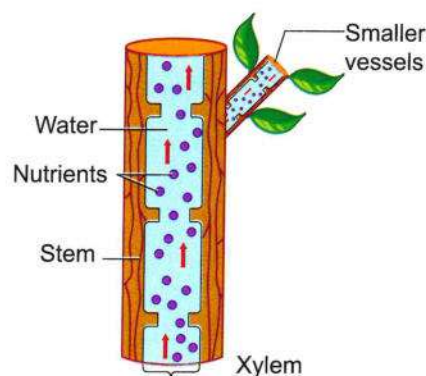
Stems

- Water and nutrients move up the plant's stem through tubes or vessels called **xylem**.

- Smaller vessels of xylem connect the stem to the leaves.

• Functions of the plant stem :

- Stem transports water and nutrients from the roots to the rest of the plant through **xylem**.
- Stem supports leaves and flowers of the plant.



features
draw
transport
vessels

بروزات
يسحب
ينقل
أوعية
root hairs
absorb
support
connect

الشعيرات الجذرية
يمتص
يدعم
يربط
anchor
increase
tubes
xylem

يثبت
يزيد
أنابيب
نسيج الخشب

► There are many forms of stems :

Forms
of
stems

Wood stem :

- Some plants have **wood stems**, such as **tree trunks** and **shrubs**.



Upright stem :

- Most **flowers** have **upright stems**.



Climb stem :

- Some plants have **climb stems**, such as **vines (grapes)**.



Tuber stem :

- Some stems extend underground and they are called **tubers**, such as **potato plant**.



Runner stem :

- Some stems run along the ground to help form new plants and they are called **runners**.



wood stem
trunk
upright stem
climb stem

ساق خشبية
جذع
ساق رأسية
ساق متسلقة
vines / grapes
shrubs
extend

نبات العنب
شجيرات
يمتد
underground
tubers
runners

تحت الأرض
الدرنات
المدادة / الجارية

Leaves

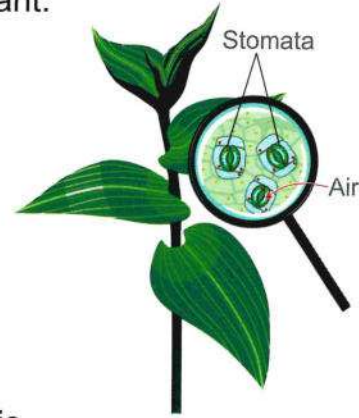
- They contain **chlorophyll**, which gives them their green color.
- **Function of the chlorophyll :**
 - Chlorophyll captures (absorbs) energy from the sunlight which allows carbon dioxide to combine with water to make food for the plant.
- The air that plant needs move into the leaves through tiny openings called **stomata**.

Stomata :

They are pores on the surface of plant's leaves that allow gases to move into and out of the plant.

• **Function of the plant leaves :**

- Leaves make food for the plant through photosynthesis process.



► There are many kinds of leaves such as :

Narrow leaves (look like needles) such as pine trees.



Flat and wide leaves.



Note

Plant's leaves get their needs of water and nutrients from the soil with the help of :

- Plant's roots.
- Xylem in the plant's stem.
- Smaller vessels of xylem connect the stem to the leaves.

capture
tiny
openings
narrow

يلتقط
صغيرة
فتحات
ضيق
pine tree
stomata
pores
needles

شجرة الصنوبر
تغور
مسام
الإبر
flat
surface
wide

مسطح
سطح
عريض

Photosynthesis process

► How does photosynthesis process occur in plant leaves ?

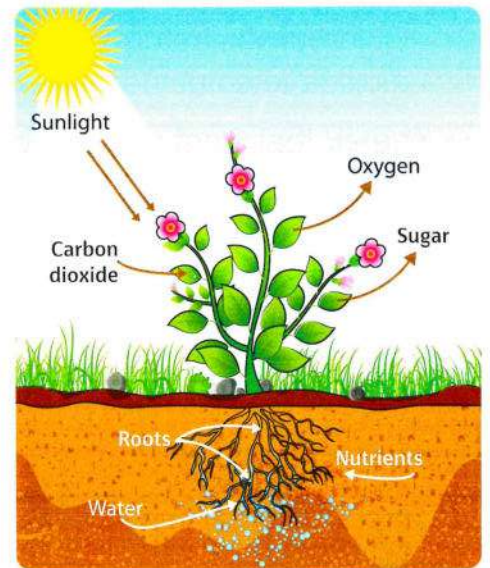
Chlorophyll absorbs energy from sunlight.

Green leaves use the light energy from the Sun to combine the carbon dioxide from the air with water.

Leaves manufacture (produce) :

- Nutrients (such as sugars, starches, fats and proteins) that the plant needs to survive.
- Oxygen gas that animals and people need to breathe.

As the photosynthesis process is completed inside the leaves, there are tubes called **phloem** that transport the food materials from the leaves to the other parts of the plant.



? Give a reason for :

The life on Earth without plants would be impossible.

Because during photosynthesis process plants produce oxygen gas that animals and people need to breathe.

Check your understanding

► Choose from column (B) what suits it in column (A) :

(A)	(B)
1. Stems	a. make food for the plant.
2. Roots	b. gives leaves their green color.
3. Leaves	c. support leaves and flowers of the plant.
4. Chlorophyll	d. fix the plant in the soil.

1.

2.

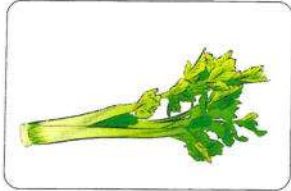
3.

4.

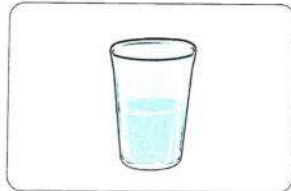
Activity 7 Up the Stem

In this activity, we will observe how the stem transports water and nutrients from the roots to all the plant parts (leaves and flowers) through **xylem vessels**.

Tools



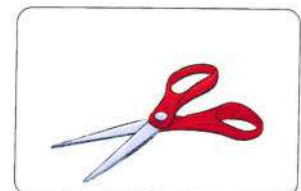
Celery stalk



Glass cup containing water



Food coloring



Scissors

Steps

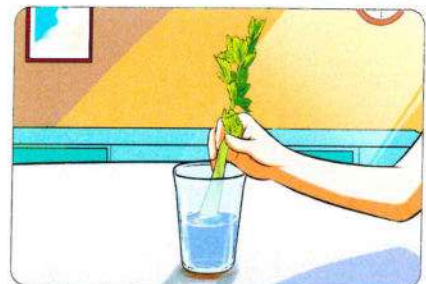
1. Fill the cup with water, then add some drops of food coloring to the water.



2. Use the scissors to cut about 2 cm off the bottom of the stalk and place it in the cup of water.



3. Leave the stalk in the water cup until the next day.

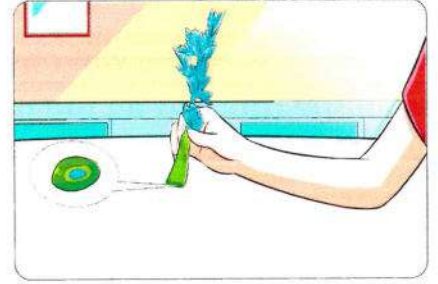


4. Cut across the celery stalk, about 5 cm up from the bottom and observe the xylem vessels inside the stalk.



► Observations

- The color of xylem will be turned into the same color of the water in the cup.
- Also, the color of leaves of celery will be turned into the same color of the water in the cup.



► Conclusion

Xylem vessels transport water and nutrients from the plant roots up to its leaves and flowers through the stem.



Check your understanding

► Put (✓) or (x) :

1. Water is transported through the xylem in the plant's stem and leaves. ()
2. Xylem helps carry water upward the plant. ()

In the Assessment Book :

Try to answer :

Self-Assessment ③

Exercises on Lesson 3

● Understand

● Apply

● Higher Thinking Skills

1 Choose the correct answer :

1. The plant's anchor it in the soil.
a. leaves b. stems c. roots d. flowers
2. There are in the plant's roots that help the plant to get more water and nutrients. *(Cairo 2024)*
a. vessels b. root hairs c. stomata d. flowers
3. The tubes that are responsible for moving water and nutrients up the plant's stem are called
a. roots. b. xylem. c. leaves. d. flowers.
4. plant has climb stem. *(Gharbia 2023)*
a. Potato b. Tomato c. Vine d. Pine
5. The kind of stems that extend underground are called stems. *(Cairo 2024)*
a. climb b. tuber c. runner d. wood
6. Potato plant has stem. *(Cairo 2024)*
a. upright b. climb c. tuber d. runner
7. Stomata are present on plant's to allow air to pass through them. *(Giza 2024)*
a. roots b. stems c. leaves d. flowers
8. can make their own food. *(South Sinai 2023)*
a. Plants only b. Animals only
c. Humans only d. Plants and some animals
9. tree has narrow leaves.
a. Potato b. Pine c. Acacia d. Grapes
10. The green plants can make their own food through
a. roots. b. stems. c. leaves. d. flowers.
11. help the plant's leaves to get water and nutrients from the soil.
a. Roots only b. Xylem only
c. Roots and xylem d. Xylem and stomata
12. All the following parts are important for plants to make photosynthesis process, except
a. roots. b. leaves.
c. stems. d. flowers.

- 13. The green color of plant's leaves is due to the presence of
a. xylem. b. phloem. c. chlorophyll. d. stomata.
- 14. Food materials are transported from the leaves to other parts of the plant through (Cairo 2023)
a. xylem. b. phloem. c. chlorophyll. d. stomata.
- 15. Animals and humans need to breathe.
a. oxygen gas b. carbon dioxide gas
c. water vapor d. sugar
- 16. Green plants produce all the following substances during photosynthesis process, except
a. oxygen gas. b. carbonyl dioxide gas.
c. starches. d. fats.

2 Choose from column (B) what suits it in column (A) :

①

(A)	(B)
1. Pine trees	a. have climb stems.
2. Potato plants	b. have runner stems.
3. Vines	c. have tuber stems.
4. Tree trunks and shrubs	d. have wood stems.
	e. have needles leaves.

1. 2. 3. 4.

②

(A)	(B)
1. Roots	a. allow gases to come in and out of the plant.
2. Stems	b. collect sunlight and carbon dioxide gas which combines with water to help the plant to make its own food.
3. Leaves	c. tubes or vessels that move water and nutrients up the plant's stem.
4. Xylem	d. absorb water and nutrients from the soil.
5. Stomata	e. transport nutrients and water from the roots to all parts of the plant.
	f. absorb oxygen gas from the soil.

1. 2. 3. 4. 5.

3 Put (✓) or (X) :

- 1. The plant is fixed in the soil by the help of its roots. (Cairo 2023) ()
- 2. Plant's stem has hairs that absorb oxygen gas from the air. (Damietta 2023) ()
- 3. Xylem helps the plant to absorb water from the soil. ()
- 4. Xylem is important for plants to transfer water from plant's roots to leaves. (Dakahlia 2023) ()
- 5. A tree trunk is a type of runner stems. ()
- 6. Potato plants have tuber stems. (Cairo 2024) ()
- 7. Vines have a kind of stems called climb stems. ()
- 8. The leaves of pine trees are flat and wide. ()
- 9. Phloem transports food materials from the leaves to other parts of the plant. (Giza 2023) ()
- 10. Photosynthesis process produces carbon dioxide gas that helps animals and humans to breathe. ()
- 11. During photosynthesis process, plant absorbs carbon dioxide gas from air through stomata. ()
- 12. There are tiny holes opening on the surface of stem that allow gases to pass into the plant. ()
- 13. Water and nutrients reach the plant's leaves with the help of roots only. ()
- 14. Plants and humans need water and air to live. ()
- 15. Plants need sunlight, oxygen gas and water to make its own food. ()
- 16. During photosynthesis process, the plant makes sugars, starches, proteins and fats that help it to survive. ()
- 17. Chlorophyll helps the plant leaves to absorb sunlight to make photosynthesis process. ()
- 18. Plants and humans are similar in the way of getting food. ()

4 Correct the underlined words :

- 1. The plant can absorb more water and nutrients from the soil by the help of xylem that are found in the roots. (Giza 2025) (.....)
- 2. There are smaller vessels of xylem that connect the root to the leaves. (.....)
- 3. Potato plant has runner stem of xylem that extends underground. (.....)
- 4. The stems that run along the ground are called tuber stems (.....)

5. Most flowers have wood stems. (.....)
6. Stomata allow water to move into and out of the plant. (.....)
7. Chlorophyll in plant's roots absorbs energy from the sunlight. (.....)
8. Animals and people can't live without carbon dioxide gas to breathe. (.....)
9. Xylem tubes transport food materials from the leaves to other parts of the plant. (.....)

5 Write the scientific term of each of the following :

1. A part of the plant that anchors it in the soil. (.....)
2. Small structures in the plant's roots that increase the absorption of water and nutrients from the soil. (.....)
3. A part of the plant that supports its leaves and flowers. (.....)
4. Vessels in plant through which water and nutrients move up from roots to leaves. (Cairo 2025) (.....)
5. The kind of plant's stem in vines. (Luxor 2025) (.....)
6. The stems that run along the ground. (.....)
7. A plant that has a tuber stem. (.....)
8. Narrow holes spread on the surface of plant's leaves that allow gases to move into and out of the plant. (Alex. 2024) (.....)
9. It is found in plant's leaves that gives them green color and absorbs energy from the sunlight. (Alex. 2023) (.....)
10. Tubes in the plant that transport food materials from the leaves to other parts of the plant. (.....)
11. The gas that the plant needs to make photosynthesis process. (.....)

6 Complete the following sentences :

1. Plant's roots the plant in the soil and absorb and water from the soil.
2. The presence of in plant's roots help it to absorb more and nutrients from the soil.
3. There are vessels called in the plant that transport water and nutrients from plant's stem to its leaves.
4. There are many kinds of stems on plants like stem in vines and stem in potato.

- 5. Shrubs have stems, while most flowers have stems.
- 6. The stems that run along the ground are called
- 7. There are tiny holes on the plant's leaves called that allow gases to move into and out of the plant. (Gharbia 2023)
- 8. Pine trees have leaves that look like needles.
- 9. Plant's leaves during photosynthesis process produce, starches, fats and that the plant needs to survive.
- 10. Food materials that are produced by process are transported from the leaves to the other parts of the plant through tubes called
- 11. The green color of plant's leaves is due to the presence of that absorbs energy from (Cairo 2024)

7 Give reasons for :

- 1. The presence of hairlike structures in plant's roots.
.....
.....
- 2. Xylem vessels are important for the plant. (Giza 2023)
.....
.....
- 3. The presence of stomata on the surface of plant's leaves.
.....
.....
- 4. Chlorophyll has an important role in photosynthesis process. (Cairo 2023)
.....
.....
- 5. There would not be life on Earth in the absence of plants.
.....
.....

8 What happens if ...?

- 1. The plant doesn't have roots. (Qena 2025)
.....
- 2. Stomata of a plant get closed for a long time.
.....
- 3. Plant's leaves don't contain chlorophyll. (Giza 2025 / Damietta 2023)
.....
- 4. The plant stops making photosynthesis process for several days.
.....

9 What is meant by (Define) :

Stomata :

10 What is the function of ... ?

1. Stomata : (Sharkia 2025)
2. Phloem : (Aswan 2025)
3. Xylem : (Menoufia 2025)
4. Chlorophyll :

11 Write one example for each of the following :

1. Wood stem :
2. Climb stem :
3. Tuber stem :
4. Upright stem :

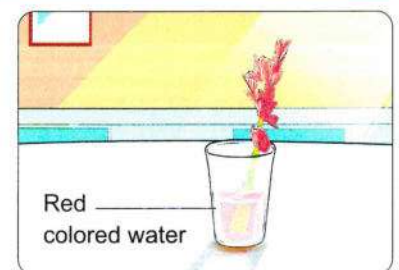
12 Complete the following comparison using these words :

(food – xylem – soil – flowers – water – photosynthesis – leaves – nutrients)

Point of comparison	Roots	Stems	Leaves
Function :	- They fix the plant in the(1)..... - They absorb(2)..... and(3)..... from the soil to the plant.	- They transport water and nutrients to the plant leaves through(4)..... - They support(5)..... and(6)..... of the plant.	- They make(7)..... for the plant through(8)..... process.

13 Look at the opposite figure, then answer :

1. The color of leaves of celery will be
2. Water is transported through that connect the stem to the leaves.



LESSON FOUR

Activity 8 Comparing Plant and Human Systems

► Put (✓) or (x) :

1. Plant needs water and air like human to survive. ()
2. Plant doesn't need energy like human to grow. ()

Need for energy

- Both plants and humans need **energy** and **gases** from the air to survive and grow as shown in the following diagrams :

Get the energy needed

Plants

- They can manufacture their own food (glucose sugar) to get energy through photosynthesis process.

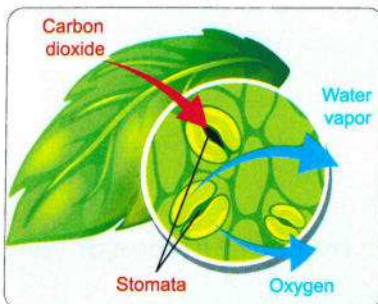
Humans

- They must eat food throughout the day to get energy, where :
 - The digestive system digests food into glucose and nutrients.
 - Then, these nutrients are absorbed into the blood.

Get the gases needed

Plants

- Air enters the plants through stomata in the leaves.



Humans

- Air enters the human body through the nose and mouth then to lungs, where oxygen is absorbed and transferred to the blood.



► **Now**, we will determine how human circulatory system is like plant transport system.

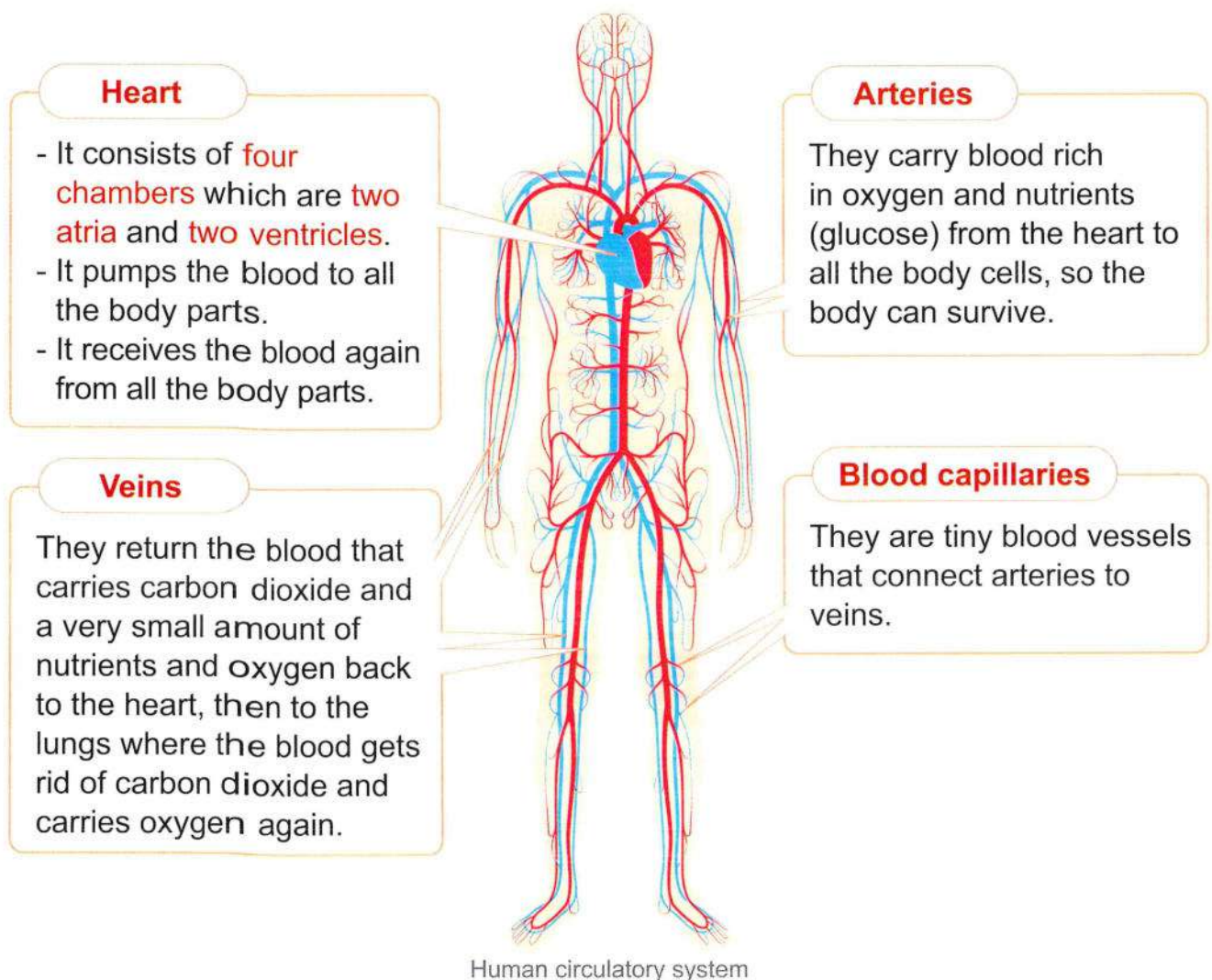
Human circulatory system

It is a system that transports oxygen and nutrients through the blood to all the body cells (parts).

Its structure :

- It consists of :

- Heart.
 - Blood vessels (tubes).
 - Blood.
- The human circulatory system has **three** different types of **blood vessels** which are :
- Arteries.
 - Veins.
 - Blood capillaries.



Note

Blood is the fluid that moves in only **one direction** in the human's arteries or veins.

circulatory system
transport
veins
ventricles

الجهاز الدوري
ينقل
أوردة
بطينان

blood vessels
arteries
cells

أوعية دموية
شريابين
خلايا

blood capillaries
pump
atria

شعيرات دموية
بضخ
أذينان

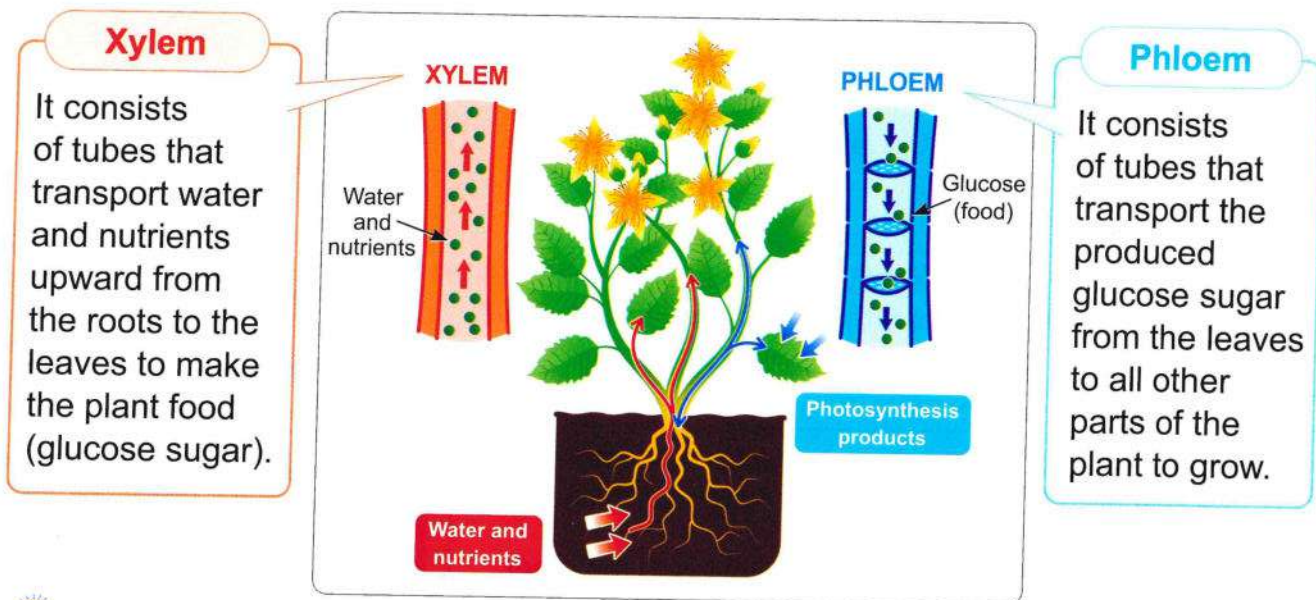
Plant transport system (Plant vascular system)

It is a system of vessels (tubes) that transports water, nutrients and plant food (glucose) between the plant parts.

Its structure :

- It consists of :

- Xylem.
- Phloem.



Note

The transport system in plants has **one-way vessels** that move important substances between the parts of the plant.

• From the previous explanation, we can conclude the similarities between the transport system in plants and circulatory system in humans, which are :

- Both have vessels to transport water, nutrients and gases.
- Both have one-way vessels (tubes).



Check your understanding

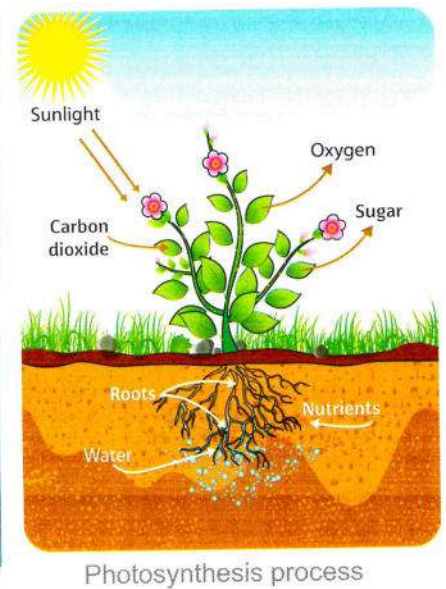
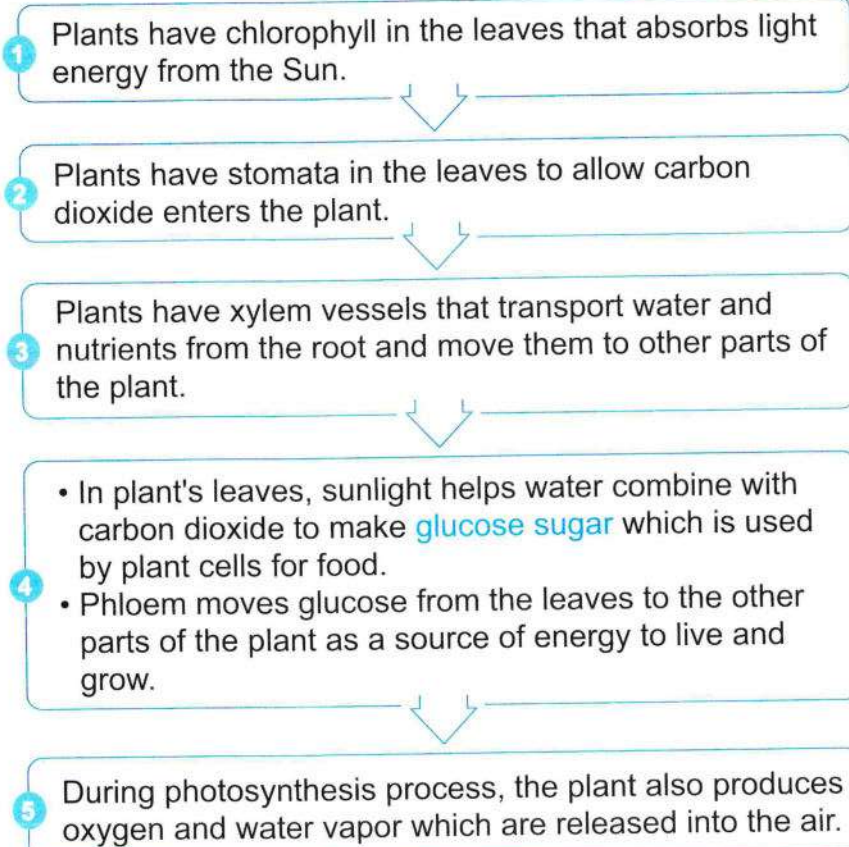
► Put (✓) or (x) :

1. Both plants and humans must take in gases from the air. ()
2. Veins carry blood rich in oxygen and nutrients. ()
3. Phloem tubes carry water to leaves. ()
4. Vessels in plants and humans are one-way vessels. ()

Activity 9 Plant Food

- Plants depend on carbon dioxide released by living organisms such as animals to make their own food during photosynthesis process.
- Also, living organisms such as animals depend on oxygen released by plants to breathe.

► We can explain the steps of photosynthesis process in plants to make their food in the following diagram :



Note

During photosynthesis process, **light energy** of the Sun is transformed into **chemical energy** that is found in glucose.



Check your understanding

► Complete the following sentences using these words :

(stomata – light – chlorophyll – chemical)

1. Plants have in their leaves to absorb sunlight.
2. Plants have in their leaves to allow carbon dioxide enters the plant.
3. During photosynthesis process, energy is transformed into energy.

Activity 10 Flowers and Seeds

Reproduction of plants

- Plants use the food they make to produce **flowers** which are responsible for **reproduction**.
- Flowers have different shapes, sizes and colors, where :
 - Some plants have large colorful flowers.
 - Some other plants, such as grasses have very small flowers.
 - There are some flowers are not very colorful.



Flowers :

They are the reproductive parts of many plants.

Function of flowers :

They produce seeds that help the plant to reproduce.

Plant reproduction :

It is the process of making new plants.

Notes

- When seeds receive **air**, **water** and **suitable temperature**, they can grow into a new plant.
- In the sunflower, the seeds are the small dark-colored objects in the center of this flower.



Check your understanding

► Put (✓) or (x) :

- In many plants, flowers are responsible for reproduction. ()
- When seeds receive air and suitable temperature only, they grow into a new plant. ()

In the Assessment Book :

Try to answer :

Self-Assessment ④

reproduction
colorful
grasses

النكاثري
ملون
الأعشاب
reproductive
seeds
suitable

نكاثري
بذور
مناسب
temperature
sunflower
responsible

درجة الحرارة
دوار / عباد الشمس
مستول

Exercises on Lesson 4

● Understand

● Apply

● Higher Thinking Skills

1 Choose the correct answer :

1. The human system that moves blood through the body is called system.
a. digestive b. respiratory c. circulatory d. nervous
2. Air enters the human body through the
a. nose only. b. mouth only.
c. nose and mouth. d. mouth and stomach.
3. The human circulatory system consists of
a. lungs, heart and blood. b. heart, blood vessels and blood.
c. blood vessels and stomach. d. heart and pancreas.
4. carry blood which is rich in oxygen and glucose from the heart to the body cells. (Dakahlia 2023)
a. Arteries b. Veins
c. Lungs and veins d. Brain and veins
5. Blood rich in carbon dioxide gas returns back to the heart through (Qena 2025)
a. arteries. b. veins. c. lungs. d. xylem.
6. system in plants consists of tubes that water and nutrients move through it.
a. Digestive b. Respiratory c. Transport d. Nervous
7. The heart in the human circulatory system consists of
a. two arteries and two ventricles. b. two atria and two ventricles.
c. two veins and two atria. d. two ventricles and two veins.
8. Glucose sugar is transported from the leaves to other parts of the plant through (Alex. 2023)
a. xylem. b. phloem. c. roots. d. stems.
9. in the leaves allow air to enter the plant. (Giza 2023)
a. Xylem b. Phloem c. Stomata d. Chlorophyll
10. In plant's leaves, light energy is converted into energy during photosynthesis process. (Alex. 2025)
a. sound b. electric c. chemical d. kinetic
11. Plants can produce new seeds by
a. roots. b. leaves. c. stems. d. flowers.

- 12. The reproductive parts of many plants are called
 a. veins. b. roots. c. leaves. d. flowers.
- 13. In , its seeds are small dark-colored objects in the center of this flower.
 a. pine tree b. sunflower c. potato plant d. celery

2 Put (✓) or (X) :

- 1. Air enters plants through their roots. (Giza 2024) ()
- 2. Living organisms need food and gases from the air to survive and grow. ()
- 3. Human circulatory system consists of the heart and the lungs. ()
- 4. Arteries are vessels in human circulatory system that carry blood rich in carbon dioxide gas. (Sharkia 2023) ()
- 5. The heart in the human circulatory system consists of two chambers. ()
- 6. Oxygen and glucose are transported from the heart to the body cells through arteries. ()
- 7. Phloem transports water and nutrients from the roots to the leaves. ()
- 8. Glucose is a type of sugar that is produced from plants during photosynthesis process. ()
- 9. The reproductive parts of many plants are flowers. (Dakahlia 2025) ()
- 10. Plant's seeds are formed inside the flowers. (Sohag 2024) ()

3 Correct the underlined words :

- 1. Human circulatory system consists of the lungs, blood vessels and blood. (.....)
- 2. Brain pumps blood to all the body parts.
- 3. Each of xylem in plants and veins in human are two-ways vessels. (.....)
- 4. Veins carry blood rich in oxygen and nutrients. (Dakahlia 2025) (.....)
- 5. During photosynthesis process, light energy is transformed into sound energy. (.....)
- 6. Plants make glucose during respiration process that provides them with energy. (.....)
- 7. Flowers of plants produce root hairs that help the plant to reproduce. (.....)

4 Write the scientific term of each of the following :

- 1. Smaller vessels that transport water and nutrients from the plant roots up through the stem to its leaves and flowers. (Cairo 2025) (.....)

- 2. The human body system that consists of the heart, blood vessels and blood. (.....)
- 3. It pumps the blood to all the body parts and receives it again. (.....)
- 4. Tiny blood vessels that connect arteries to veins. (.....)
- 5. A system of tubes through which water, nutrients and plant food are carried all over the plant. (Cairo 2024) (.....)
- 6. Blood vessels carry blood from the heart to all the body parts. (.....)
- 7. Blood vessels carry blood from the body parts and return it back to the heart. (Giza 2025) (.....)
- 8. A type of sugar produced by the plant during photosynthesis process. (Suez 2025) (.....)
- 9. Vessels move glucose from the leaves to other parts of the plant. (.....)
- 10. Parts of the plant that are responsible for reproduction. (.....)
- 11. The process of producing new plants. (.....)
- 12. The source of energy that the plant uses to make photosynthesis process. (.....)

5 Complete the following sentences :

- 1. Plants make their food in the form of sugar during photosynthesis process.
- 2. Air enters plants through stomata on their , while it enters the human body through and
- 3. Human circulatory system consists of , and
- 4. Arteries carry blood rich in and from the heart to all the body parts. (Gharbia 2024)
- 5. The nutrients and oxygen are transported through the human blood to the body cells by the system. (Alex. 2024)
- 6. The heart in the human circulatory system consists of and
- 7. The plant makes sugar in its during photosynthesis process.
- 8. Transport system in the plant consists of two types of vessels which are and
- 9. Arteries carry oxygen and nutrients from the to all the body parts, while in plant's stem carries water from the to the leaves.
- 10. In plant's leaves, energy is converted into energy during photosynthesis process. (Menoufia 2023)
- 11. Flowers of the plant produce that help it to
- 12. There are three types of vessels in the human circulatory system which are , and (Cairo 2023)

6 Give reasons for :

- 1. Xylem in plant is a one-way vessel.

.....

.....

- 2. Flowers are important parts for the plant.

.....

.....

7 What happens if ...?

- 1. Plants can't produce glucose sugar during photosynthesis process.

.....

.....

- 2. We remove the flowers of a plant.

(Menoufia 2023)

.....

.....

8 What is meant by (Define) :

- 1. Flowers :

- 2. Plant reproduction :

9 What is the function of ... ?

- 1. Human circulatory system :

- 2. Heart :

- 3. Arteries :

- 4. Veins :

- 5. Plant transport system :

- 6. Flowers :

10 Complete the following comparison using the words below :

(xylem – veins – plant parts – blood – phloem – arteries)

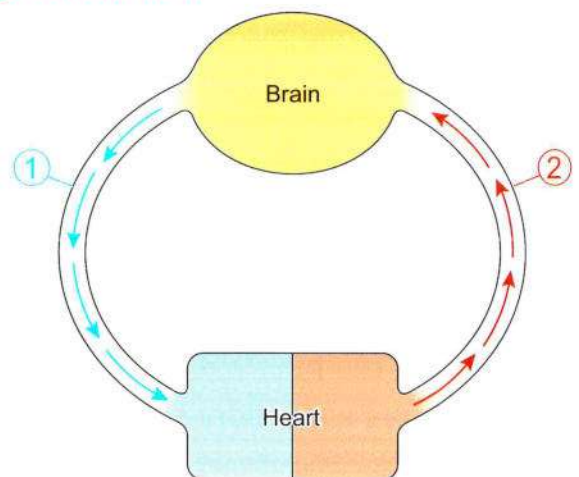
Plant transport system	Human circulatory system
- It transports different materials around the(1).....	- It transports the(2)..... around the human body.
- Water and nutrients are carried from the roots to the leaves through(3)..... tubes.	- Blood rich in oxygen and nutrients is carried from the heart to all the body parts through(4).....
- Glucose sugar is carried from the leaves to all the plant parts through(5).....	- Blood rich in carbon dioxide is carried from all the body parts to the heart through(6).....

11 Arrange the following sentences to describe the process that converts energy from the Sun into food inside the plant in the correct order :

- (.....) Vessels move glucose from the leaves to other parts of the plant.
- (.....) Light from the Sun hits plant's leaves.
- (.....) Plant parts use the glucose for their needs and growth.
- (.....) The leaves transform light energy from the Sun into glucose (chemical energy).

12 Look at the opposite figure, then choose the correct answer from those between brackets to complete the following sentences :

1. Vessel number ① refers to
(a vein – an artery)
2. Vessel number ② refers to
(a vein – an artery)
3. Vessel number (① – ②)
transfers blood rich in carbon dioxide gas,
while vessel number (① – ②)
transfers blood rich in oxygen gas.
4. Vessels number ① and ② belong to
..... system. (digestive – circulatory)



LESSON FIVE

Activity 11 Seed Dispersal

► Put (✓) or (x) :

1. Plants use the energy they get from food they make to produce seeds. ()
 2. Flowers produce seeds for the plant to help it to reproduce. ()
- Seeds are transported from one place to another, this process is called **seed dispersal**.

Ways of seed dispersal in nature

[1] Water :

Seeds that are dispersed by **water** can **float** on water.

Example: Coconut seeds.



[2] Wind :

Seeds that are dispersed by **wind** are **light**.

Examples:

- Maple seeds.
- Dandelion seeds.



[3] Animals or human transport :

Seeds that are dispersed by **animals or human transport** can **stick** to animal fur or human clothes.

Example: Burdock seeds (have spines).



[4] Seeds that are eaten by animals :

Some seeds can be dispersed when they **come out with the animals' stool** in another place.

Examples:

- Tomato seeds.
- Apple seeds.



Note

Different ways of seed dispersal depend on the different properties of seeds (such as : size, shape , etc).



Check your understanding

► Put (✓) or (x) :

1. Light seeds travel in the air. ()
2. Seeds with spines stick to animal fur. ()

nature

الطبيعة

light

خفيفة

spines

أشواك

maple seeds

بذور القيقب

float

يطفو

stick

يلصق / يعلق

stool

البراز

burdock seeds

بذور الأرقطيون

dandelion seeds بذور الهندباء

fur

فراء

coconut seeds بذور جوز الهند

Activity 12 Record Evidence Like A Scientist

You have learned a lot about plant needs and plant structures.

In this activity, which will be repeated at the end of each concept, we will learn how to think like scientists to answer a question about one of the main points of this concept through four main steps :

- **Step ①** : The Question.
- **Step ②** : My Claim.
- **Step ③** : My Evidence.
- **Step ④** : My Scientific Explanation.

? Step 1 The Question

How do the structures of a plant use water, air and light to perform life processes ?

💡 Step 2 My Claim

- Plants use different parts to obtain their basic needs of water, air and light to make their own food through photosynthesis.
- Each part of a plant has a function to help it survive.



Note

Your claim should be formed of a sentence that gives an answer for the previous question in step ①.

🔍 Step 3 My Evidence

- In most plants, the roots absorb water and nutrients from the soil, and then the stem moves the water up to the leaves.
- If a green plant is placed in a dark place for many days, their leaves will turn yellow and the plant will die, so green plant needs sunlight to survive.



Note

You should mention enough and suitable evidence that support your claim.



Step 4 My Scientific Explanation

- Plant roots absorb water and nutrients from the soil, then the stem transports them to the leaves through xylem.
- Plant leaves absorb carbon dioxide from air through stomata and absorb the sunlight through chlorophyll.
- During photosynthesis process, green leaves use the light energy from the Sun to combine the carbon dioxide from the air with water to produce glucose sugar (plant's food) and oxygen gas that all living organisms need to breathe.



Note

Your scientific explanation should explain your claim and evidence introducing some supportive examples from what you have learned.

Review on Concept (1 - 1)

To review this concept look at the **Assessment Book** "Part 2 : Final Revision".

In the Assessment Book :

Try to answer :

- Self-Assessment ⑤
- Model Exam on Concept (1.1)

Exercises on Lesson 5

● Understand

● Apply

● Higher Thinking Skills

1 Choose the correct answer :

- 1. The movement of seeds from a place to another is called (Qena 2025)
 - a. seeds germination.
 - b. seeds dispersal.
 - c. seeds reproduction.
 - d. seeds growth.
- 2. All the following can help in seed dispersal, except
 - a. wind.
 - b. water.
 - c. human and animals.
 - d. soil and sunlight.
- 3. Maple seeds travel by wind because they are (Cairo 2023)
 - a. light seeds.
 - b. spiny seeds.
 - c. heavy seeds.
 - d. smooth seeds.
- 4. Burdock seeds have spines, so they can
 - a. float on water.
 - b. travel by wind.
 - c. stick to animal fur.
 - d. be eaten by animals.
- 5. From the ways of seeds dispersal is floating on water as in (Suez 2023)
 - a. burdock seeds.
 - b. tomato seeds.
 - c. dandelion seeds.
 - d. coconut seeds.

2 Choose from column (B) what suits it in column (A) :

(A)	(B)
1. Coconut seeds	a. stick to animal fur.
2. Maple seeds and dandelion seeds	b. float on water.
3. Burdock seeds	c. are eaten by animals.
4. Tomato seeds and apple seeds	d. travel by wind.
	e. stay inside flowers without movement.

1. 2. 3. 4.

3 Put (✓) or (X) :

- 1. Seeds germination means the transportation of seeds from one place to another. ()
- 2. There are many ways of seeds dispersal in nature. ()
- 3. Coconut seeds can float on water. (Beheira 2024) ()
- 4. Dandelion seeds have spines, so they stick to animal fur. ()
- 5. Tomato seeds are light so they can disperse through air. (Beheira 2023) ()
- 6. Human could be one of the ways of seeds dispersal. ()

4 Correct the underlined words :

1. Coconut seeds are dispersed by wind. (Minia 2023) (.....)
2. Burdock seeds are light seeds. (Aswan 2023) (.....)
3. Tomato and coconut seeds are eaten by animals and come out with their stool. (.....)

5 Write the scientific term of the following :

- Transferring of seeds from a place to another. (Beheira 2025) (.....)

6 Complete the following sentences :

1. Some seeds can be transported from one place to another by floating on water as seeds or traveling by wind as seeds. (Qalyoubia 2023)
2. Burdock seeds can stick to animal fur because they have
3. Maple seeds and dandelion seeds can travel by wind because they are

7 Give reasons for :

1. Seeds dispersal may take place by animal in two different ways.

.....

.....

2. Seeds of maple or dandelion plants can disperse through wind easily.

(Sharkia 2025 / Fayoum 2023)

.....

.....

3. Burdock seeds can stick to animal fur.

(Cairo 2024)

.....

.....

1

Total mark

15

(5 marks)

-
-

(5 marks)

-

3 (A) Write the scientific term of each of the following :**(5 marks)**

1. A liquid substance that plants, animals and humans need to survive. (.....)
2. Parts of the plant that are responsible for reproduction. (.....)
3. The source of energy for the plant to make photosynthesis process. (.....)
4. The plant that has a tuber stem. (.....)

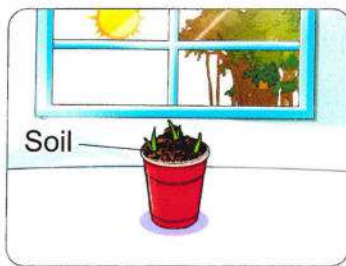
(B) Look at the following figures, then complete the following sentences using the words below :**(soil – figure (A) – figure (B))**

Figure (A)



Figure (B)

1. The seeds in grow faster than those in
2. Seeds in figure (B) should be transferred into to complete its growth.

1 (A) Complete the following sentences :

(5 marks)

1. There are small vessels that transfer and nutrients from the plant's stem to leaves.
2. In plant's leaves, light energy of the Sun is converted into energy during photosynthesis process.
3. Arteries carry oxygen and nutrients from the to all the body parts.
4. Tree trunks have stems.

(B) Give a reason for the following :

There is no life on Earth in the absence of plants.

.....

.....

2 (A) Choose from column (B) what suits it in column (A) :

(5 marks)

(A)	(B)
1. Roots	a. allow gases to move into and out of the plant.
2. Stems	b. collect sunlight and carbon dioxide gas which combines with water to help the plant to make its own food.
3. Leaves	c. absorb water and nutrients from the soil.
4. Stomata	d. transport water and nutrients from the roots to all parts of the plant.
	e. absorbs oxygen gas from the soil.

1. 2. 3. 4.

(B) Correct the underlined words :

1. Chlorophyll in plant's roots absorbs energy from the sunlight. (.....)
2. Phloem tubes carry water and nutrients from the roots to the leaves. (.....)

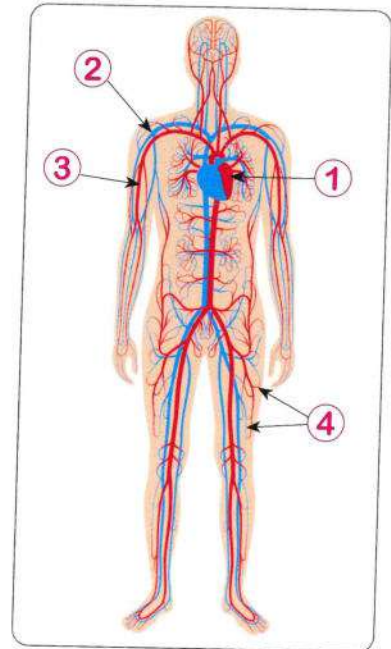
3 (A) Choose the correct answer :**(5 marks)**

1. tree has narrow leaves.
 a. Potato b. Pine c. Acacia d. Grapes
2. Plants can produce new seeds by
 a. roots. b. leaves. c. stems. d. flowers.
3. seeds travel by wind.
 a. Coconut b. Maple c. Burdock d. Apple
4. The heart in the human circulatory system consists of
 a. two arteries and two ventricles.
 b. two atria and two ventricles.
 c. two veins and two atria.
 d. two ventricles and two veins.

(B) Look at the opposite figure, then answer :

1. The opposite figure represents the human system.
2. Label the figure :

- ①
- ②
- ③
- ④

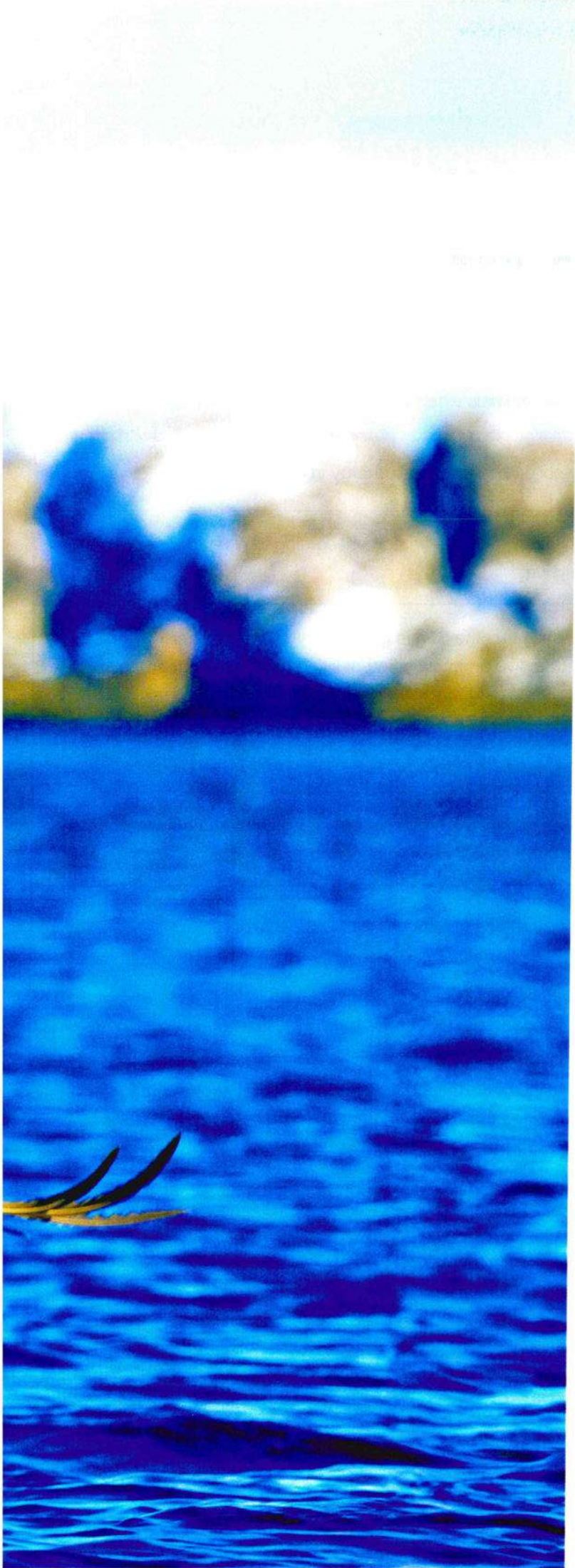


CONCEPT

1.2

Energy Flow in Ecosystems





Learning outcomes

By the end of this concept, your child will be able to :

- Develop a model to show how energy moves through an ecosystem.
- Create a model to explain the different roles that organisms play in an ecosystem.
- Explain how the health of each type of organism in an ecosystem impacts the overall health of the community.

Key vocabulary

- | | |
|-------------|---------------|
| • Consumers | • Decomposers |
| • Ecosystem | • Food chain |
| • Food web | • Interact |
| • Predators | • Prey |
| • Producers | |

Notes For Parents On Concept [1.2]

Lessons	Activities	What you should do with your child
1	Activity 1	Explain to your child how the energy flows through an ecosystem from plants to animals and between animals when they eat each other.
	Activity 2	Discuss with your child how hawk gets energy in an ecosystem.
	Activity 3	Explain to your child how animals eat food according to what these animals bodies need to survive.
2	Activity 4	Discuss with your child that the Sun is the primary source of energy for all organisms on Earth to live and how different living organisms get energy.
	Activity 5	Explain to your child that living organisms can be classified into three groups according to their way of feeding.
	Activity 6	Discuss with your child how the movement of energy and nutrients through an ecosystem can be represented using model known as a food chain.
3	Activity 7	Let your child make a model of a food chain.
	Activity 8	Explain to your child how all living organisms interact in food webs and we can draw these webs to show how organisms are connected within ecosystem.
	Activity 9	Discuss with your child how the food web is a model that shows many interactions among living organisms in an ecosystem.
4	Activity 10	Help your child to think like a scientist by answering a question about one of the main points of this concept, then write his/her claim, evidence and the scientific explanation.
	Activity 11	Discuss with your child how restoration ecology is very important for plants and animals that help them have a stable environment to survive.